



MediSoft HBYS

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Hospital Management System

Microservices Architecture

*A Polyglot Microservices Hospital
Information System*

Spring
Boot

Spark
Java

Dropwizar
d

Vert.
x

Reac
t

Agenda

What we will cover today

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What is MediSoft HBYS?

A Polyglot Microservices Hospital Information System



Goal

Simulate a real Turkish HBYS
(Hastane Bilgi Yönetim Sistemi)



Approach

5 microservices, each built
with a different Java framework



Deployment

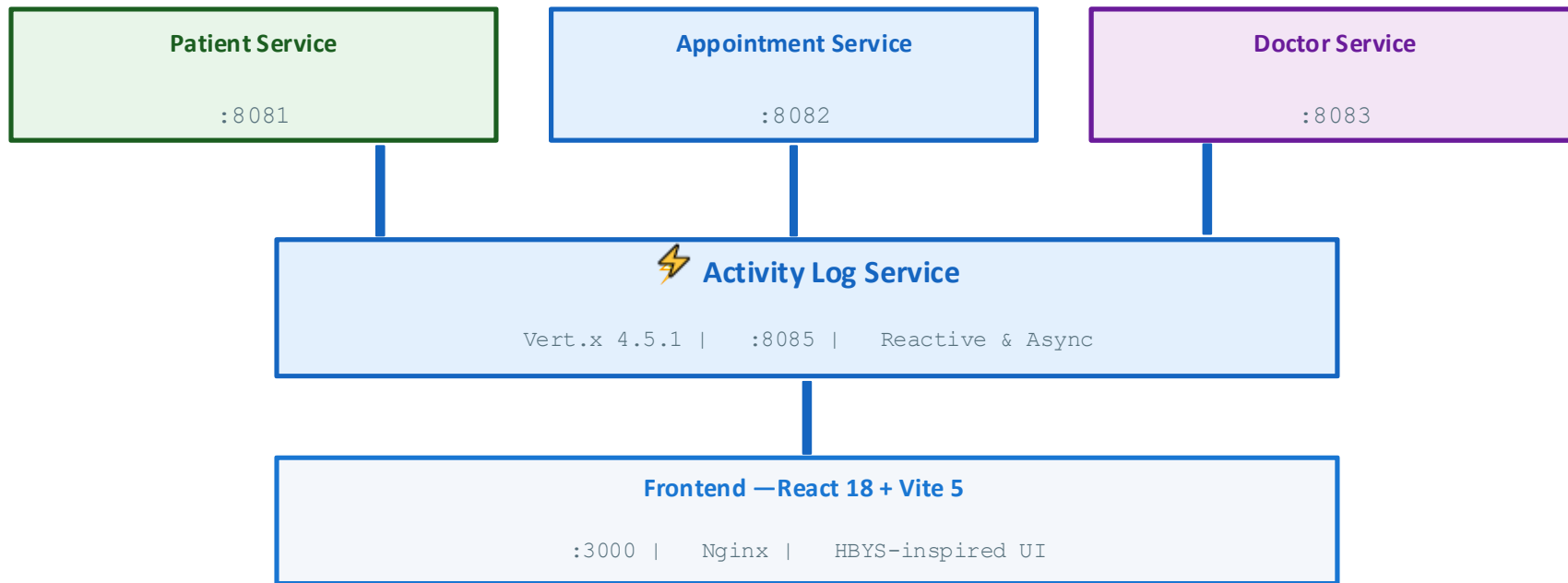
Fully containerized with
Docker Compose —one command



MediSoft HBYS —Merkez Hastane

System Architecture

5 containers — 1 Docker bridge network — hospital-network



```
docker-compose up --build
```

Five Services at a Glance

Polyglot — each service uses a different technology stack

Patient Service	Spring Boot	17 (LTS)	8081	H2 Database (JPA)
Appointment	3.2.0 Spark Java	11 (LTS)	8082	HashMap (In-memory)
Service Doctor	2.9.4	11 (LTS)	8083	HashMap (In-memory)
Service Activity Log	Dropwizard	11 (LTS)	8085	ArrayList (In-memory)
Service Front end	2.1.6 Vert.x 4.5.1 React 18 + Vite 5	Node 20	3000	— (SPA)

Each service is independently buildable, deployable, and scalable —the core principle of microservices.

Patient Service —Spring Boot 3.2.0

Java17LTS | H2In-memoryDatabase | Port8081

Framework	Spring Boot 3.2.0
Java	17 (LTS)
Database	H2 + Spring Data JPA
Port	8081
Build	spring-boot-maven-plugin
CORS	WebMvcConfigurer

GET	/api/patients
GET	/api/patients/{id}
GET	/api/patients/tc/{tcNo}
POST	/api/patients
PUT	/api/patients/{id}
DELETE	/api/patients/{id}

Appointment Service —Spark Java 2.9.4

Lightweight micro-framework — entire service in ONE Java file

Spark Java: Express-style DSL | Embedded Jetty | No boilerplate | Port 8082

Framework	Spark Java 2.9.4
Java	11 (LTS)
JSON	Gson 2.10.1
Port	8082
Storage	HashMap + AtomicLong

```
port(8082)
;
before((req, res) ->
{
  res.header("Access-Control-Allow-Origin",
  })  "*");
;
get("/api/appointments", (req, res) -
>  gson.toJson(new
    ArrayList<>{appointments.values()}));
post("/api/appointments", (req, res) ->
{
  Appointment a = gson.fromJson(req.body(),
  Appointment.class); a.id = idGenerator.getAndIncrement();
  return gson.toJson(a);
})
;
```

Appointment Status:

BEKLIYOR

TAMAMLANDI

IPTAL

Doctor Service — Dropwizard 2.1.6

JerseyJAX-RS+EmbeddedJetty+Jackson | Port8083/8084

Framework	Dropwizard 2.1.6
Java	11 (LTS)
REST	Jersey JAX-RS
Port (App)	8083
Port (Admin)	8084
CORS	CrossOriginFilter

Preloaded Doctor Registry

Dr. Ahmet	Kar di yol	✓ Ye
Yilmaz Dr. Ayse	oji Nor ol	✓ s
Demir Dr.	oji	X Ye
Mehmet Kaya	Ortopedi	✓ s
Dr. Fatma Celik	Dahiliye	No
		Ye
		s

REST Endpoints:

GET /api/doctors

GET /api/doctors/available

GET

/api/doctors/specialty/{spec}

POST /api/doctors

PUT /api/doctors/{id}

DELETE}

/api/doctors/{id}

Activity Log Service —Eclipse Vert.x 4.5.1

Reactive&Asynchronous | Port8085 | Centralizedlogginghub



Reactive Event-Loop Model —Non-blocking I/O —Fire-and-Forget



Event-Loop

Like Node.js for JVM.
No thread-per-request. Memory efficient.



Central Hub

All 3 services send logs here.
Unified log storage.



Non-blocking I/O

Handles burst traffic without thread pool exhaustion.



Fire-and-Forget

Senders use `sendAsync()` —no waiting for response.

Endpoints:

GET

/logs

POST

/logs

DELETE

/logs

GET

/health

Log Levels:

INFO

WARN

ERROR

DEBUG

Frontend —React 18 + Vite 5

HBYS-inspiredUI | Nginx | Port3000 | MediSoftHBYS



Ana Ekran
Dashboard

Stats cards, recent patients & appointments, doctor status



Hasta Kayıt
Patient Mgmt

CRUD, protocol P-000001
TC No, blood type
gender



Poliklinik Randevusu
Appointments

Calendar date picker
Hourly time 08:00–18:00
Status management



Hekim Listesi
Doctor Mgmt

Specialty filter
Availability toggle
Uzm. Dr. prefix



Aktivite Logları
Activity Logs

Real-time auto-refresh 3s
●CANLI live indicator
Filter by service/level

UI Features:

Real-time Clock

Dark Sidebar

Color-coded Badges

●CANLI Indicator

P-000001 Protocol

Multi-stagebuilds | docker-compose.yml | hospital-network

activity-log-service	: 8085
patient-service	: 8081
appointment-service	: 8082
doctor-service	:8083 / :8084
frontend	: 3000

Startup order via depends_on: activity-log-service → patient + doctor → appointment → frontend

Multi-Stage Dockerfile Pattern

```
# Stage 1 -Build
FROM maven:3.9.5-eclipse-temurin-11 AS
build WORKDIR /app
COPY pom.xml .
COPY src ./src
RUN mvn clean package -DskipTests

# Stage 2 -Runtime (slim image)
FROM eclipse-temurin:11-jre
COPY --from=build /app/target/*.jar
app.jar
EXPOSE 8085
ENTRYPOINT ["java", "-jar", "app.jar"]
```

```
docker-compose up --build -d
```

Inter-Service Communication

Async Fire-and-Forget Logging Pattern

```
// Same pattern in ALL 3 business services
private static void sendLog(String message, String level)
{
    try
    {
        String body =
            String.format(
                "{\"service\":\"...\",\"message\":\"%s\",\"level\":\"%s\"}",
                message, level);
        HttpRequest req =
            HttpRequest.newBuilder()
                .uri(URI.create(LOG_URL))
                .header("Content-Type", "application/json")
                .POST(HttpRequest.BodyPublishers.ofString(body)).build();
        // NON-BLOCKING -does not wait for response
        httpClient.sendAsync(req,
            HttpResponse.BodyHandlers.ofString());
    } catch (Exception ignored) {} // Never crashes main
}
service
```

Patient Service	Patient created	INFO
Patient Service	Patient deleted	WAR
Appointment Service	Appointment created	N INFO
Appointment Service	Appointment deleted	WAR N
Doctor Service	Doctor added	INFO
Doctor Service	Doctor deleted	WAR

N

If the Activity Log Service goes down —the other services keep running normally.

CORS —Cross-Origin Resource Sharing

Why it's needed and how each framework handles it

Frontend :3000 → calls APIs at :8081, :8082, :8083, :8085 → Browser blocks by default (Same-Origin Policy)

Spring	WebMvcConfigurer.addCorsMappings	CorsConfig.java @Configuration
Boot Spark	() before() filter —sets headers	Appointment Service. main()
Java	CrossOriginFilter servlet filter	DoctorApplication.run()
Dropwizar	Manual putHeader() in route handler	ActivityLogVerticle.start()

d Vert.x

```
// All services respond with:  
Access-Control-Allow-Origin: * Access-Control-Allow-  
Methods: GET, POST, PUT, DELETE, OPTIONS Access-Control-  
Allow-Headers: Content-Type, Authorization
```

Technologies Used

Complete stack — backend, frontend, and DevOps

BACKEND

Spring Boot	3.2.0	Patient Service REST + JPA
Spark Java	2.9.4	Appointment lightweight DSL
Dropwizard	2.1.6	Doctor Service JAX-RS
Vert.x	4.5.1	Log Service reactive HTTP
Java 17	LTS	Patient Service JVM
Java 11	LTS	All other Java services
H2 Database	embed	Patient persistence
Gson	2.10.1	JSON (Appointment Service)

FRONTEND & DEVOPS

React	18	Frontend SPA framework
Vite	5	Build tool + HMR
Nginx	latest	Static file serving
Docker	latest	Containerization
Docker Compose	latest	Multi-container orchestration
Node	20	Frontend build environment

Key Takeaways

What this project demonstrates



Polyglot Microservices

Spring Boot, Spark Java, Dropwizard, Vert.x —the right tool for each specific job.



Reactive Programming

Vert.x event-loop model handles burst logging traffic without thread pool exhaustion.



Docker Orchestration

`docker-compose up --build` deploys all 5 services in the correct order with one command.



Async Communication

`sendAsync()` fire-and-forget: inter-service logging never blocks or breaks business logic.



HBYS-Inspired UI

Realistic Turkish hospital system look inspired by Probel HBYS —MediSoft HBYS branding.



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***THANK YOU FOR
LISTENING***

ANY QUESTIONS ?